

MySQL Database Schema Specification

Data Model & Field Specifications for Retail & Customer Analytics Assignment

Assignment Instructions for Students

Analyze the 6 CSV data descriptions below. Create the target relational database schema in MySQL using proper Primary Keys (PK), Foreign Keys (FK), index constraints, and optimal column data types (`INT`, `VARCHAR`, `DECIMAL`, `DATE`, `DATETIME`).

1. customers

CSV Source: customers.csv

Demographic and basic registration information for all registered customers.

COLUMN NAME	SUGGESTED MYSQL TYPE	DESCRIPTION & CONSTRAINTS
customer_id PK	VARCHAR(50) / INT	Unique identifier for each customer. Primary Key.
full_name	VARCHAR(100)	Full name of the customer.
age	INT	Age of the customer in years.
gender	VARCHAR(20)	Gender identity (e.g., 'Male', 'Female', 'Other').
email	VARCHAR(150)	Primary email address.
phone	VARCHAR(30)	Contact phone number.
street_address	VARCHAR(255)	Street address details.
city	VARCHAR(100)	City of residence.
state	VARCHAR(100)	State / Province of residence.
zip_code	VARCHAR(20)	Postal or zip code.
registration_date	DATE	Date when the account was registered.
preferred_channel	VARCHAR(30)	Preferred communication channel (e.g., 'Email', 'SMS').

2. transactions

[CSV Source: transactions.csv](#)

Purchase records detailing items bought by customers across stores and online.

COLUMN NAME	SUGGESTED MYSQL TYPE	DESCRIPTION & CONSTRAINTS
transaction_id PK	INT AUTO_INCREMENT	Unique transaction identifier. Primary Key.
customer_id FK	VARCHAR(50) / INT	Foreign Key referencing customers(customer_id).
product_name	VARCHAR(150)	Name/description of the product purchased.
product_category	VARCHAR(80)	Category classification of the product.
quantity	INT	Quantity of units purchased.
price	DECIMAL(10,2)	Unit price of the product.
transaction_date	DATE	Date when the transaction took place.
store_location	VARCHAR(100)	Physical store location or 'Online'.
payment_method	VARCHAR(50)	Payment method used (Credit Card, PayPal, Cash).
discount_applied	DECIMAL(5,2)	Discount amount or percentage applied.

3. interactions

[CSV Source: interactions.csv](#)

Digital activity logs capturing customer behavior on web, app, and social channels.

COLUMN NAME	SUGGESTED MYSQL TYPE	DESCRIPTION & CONSTRAINTS
interaction_id PK	INT AUTO_INCREMENT	Unique interaction log identifier. Primary Key.
customer_id FK	VARCHAR(50) / INT	Foreign Key referencing customers(customer_id).
channel	VARCHAR(50)	Channel used (Website, Mobile App, Social Media).
interaction_type	VARCHAR(50)	Action taken (View Product, Add to Cart, Click Ad).
interaction_date	DATETIME / DATE	Date and timestamp of interaction.
duration	DECIMAL(8,2)	Duration of activity in seconds or minutes.
page_or_product	VARCHAR(150)	Page URL visited or product name referenced.
session_id	VARCHAR(100)	Unique session ID tracking consecutive actions.

4. campaigns

[CSV Source: campaigns.csv](#)

Marketing and advertising campaign performance metrics and budget details.

COLUMN NAME	SUGGESTED MYSQL TYPE	DESCRIPTION & CONSTRAINTS
campaign_id PK	INT AUTO_INCREMENT	Unique campaign identifier. Primary Key.
campaign_name	VARCHAR(150)	Title or campaign name.
campaign_type	VARCHAR(50)	Medium used (Email, Social Media, TV Ad).
start_date	DATE	Campaign launch date.
end_date	DATE	Campaign completion date.
target_segment	VARCHAR(100)	Target audience (Loyal Customers, New Registrations).
budget	DECIMAL(12,2)	Allocated monetary budget.
impressions	INT	Total ad impressions generated.
clicks	INT	Total ad clicks recorded.
conversions	INT	Number of completed target actions/conversions.
conversion_rate	DECIMAL(5,4)	Ratio/percentage of conversions per impression.
roi	DECIMAL(8,2)	Return on Investment calculation.

5. customer_reviews_complete

[CSV Source: customer_reviews_complete.csv](#)

Product ratings and written feedback provided by verified purchasers.

COLUMN NAME	SUGGESTED MYSQL TYPE	DESCRIPTION & CONSTRAINTS
review_id PK	INT AUTO_INCREMENT	Unique review identifier. Primary Key.
customer_id FK	VARCHAR(50) / INT	Foreign Key referencing customers(customer_id).
product_name	VARCHAR(150)	Name of product reviewed.
product_category	VARCHAR(80)	Product category.
full_name	VARCHAR(100)	Customer display name.
transaction_date	DATE	Associated transaction purchase date.
review_date	DATE	Date review was posted.
rating	INT	Star rating (1 to 5).
review_title	VARCHAR(255)	Short summary title of the review.
review_text	TEXT	Full narrative customer feedback text.

6. support_tickets

CSV Source: [support_tickets.csv](#)

Customer service ticket tracking, resolution timing, and satisfaction metrics.

COLUMN NAME	SUGGESTED MYSQL TYPE	DESCRIPTION & CONSTRAINTS
ticket_id PK	INT AUTO_INCREMENT	Unique support ticket identifier. Primary Key.
customer_id FK	VARCHAR(50) / INT	Foreign Key referencing customers (customer_id).
issue_category	VARCHAR(80)	Issue type ('Billing', 'Product Defect', 'Returns').
priority	VARCHAR(20)	Ticket priority level ('Low', 'Medium', 'High').
submission_date	DATETIME / DATE	Date and time ticket was logged.
resolution_date	DATETIME / DATE	Date and time ticket was resolved.
resolution_status	VARCHAR(30)	Current status ('Open', 'Closed', 'Pending').
resolution_time_hours	DECIMAL(8,2)	Elapsed time to resolution in hours.
customer_satisfaction_score	DECIMAL(3,2) / INT	Post-resolution CSAT score (1–5 scale).
notes	TEXT	Agent notes and resolution summary details.